

supervisor	
signature	

* You should write your answer in English. Otherwise, penalty may be applied.

- (a.) is a term used in the early days of computing science for the difficulty of writing correct, useful and efficient computer programs in the required time. The (same as (a)) was due to the rapid increases in computer power and the (b.) of the problems that could not be tackled. With the increase in the (same as (b)) of the software, many software problems arose because existing methods were inadequate.

- In C++, the file open mode (`ios::trunc`) means that if the file is opened for output operations and the file is already (e. g. opened), its previous content is deleted and replaced by the new one.

- (f.) is the process of changing, modifying, and updating software to keep up with customer needs. (same as (f)) is done after the product has launched for several reasons including improving the software overall, correcting issues or bugs, to boost performance, and more.

- Coupling is the degree to which each program module (g. M_i) depends on each one of the other modules. (h. C_i) is a measure of how strongly-related and focused the various responsibilities of a software module are. It is often desirable to (i. C_i) coupling and (j. C_i) (same as (h)).

- One of the main benefits of data encapsulation is (k.). (same as (k)) means (l.) in implementation should not affect (m.) seen by clients of a class.

- (n.) member function does not (o.) any data members of the object on which it is called. This restriction is enforced by (p.). (same as (n)) member function assures its clients of its benign natures (i.e. safety and readability).

- example of static memory allocation for using an array

```
int x[10];
x[2]=7;
```
- example of dynamic memory allocation for using an array

```
int* x = new int[10];
x[2]=7;
```

Describe the main advantage and disadvantage of static memory allocation and dynamic memory allocation for using an array by filling in the blanks in the following table. You also have to explain why static memory allocation and dynamic memory allocation can get such advantage and disadvantage in the following table.

	main advantage and why such advantage can be obtained?	main disadvantage and why such disadvantage can be obtained?
static memory allocation for using array	advantage:(<u>use less than 10 English words</u>) why?: (<u>use less than 15 English words</u>)	disadvantage:(<u>use less than 10 English words</u>) why?: (<u>use less than 15 English words</u>)
dynamic memory allocation for using array	advantage:(<u>use less than 10 English words</u>) why?: (<u>use less than 15 English words</u>)	disadvantage:(<u>use less than 10 English words</u>) why?: (<u>use less than 15 English words</u>)

3.(20 points) Consider following C++ code and its area calculation screen output. Fill in empty boxes with appropriate C++ code.

```

1: #include <iostream>
2: using namespace std;
3: class Rectangle {
4:     int width, height;
5:     [ ]
6:     Rectangle() { width=1; height=1; }
7:     Rectangle(int x, int y) [ ] {}
8:     int area() { return width*height; }
9: };
10: int main() {
11:     Rectangle obj(3,4);
12:     Rectangle *foo, [ ];
13:     foo = [ ];
14:     bar = new Rectangle (5,6);
15:     baz = new Rectangle[2];
16:     cout << "Area Calculation: ";
17:     cout << "obj:" << obj [ ] << ", ";
18:     cout << "foo:" << foo [ ] << ", ";
19:     cout << "bar:" << bar [ ] << ", ";
20:     cout << "baz[0]:"<< baz[0] [ ] << "\n";
21:     delete bar; delete[] baz; return 0;
22: }

===== screen output =====
Area Calculation: obj:12, foo:12, bar:30, baz[0]:1

```

4.(8 points) Consider following C++ code and its screen output showing the size of input file "example.bin". Fill in empty boxes with appropriate C++ code.

```

1: #include <iostream>
2: #include <fstream>
3: using namespace std;
4:
5: int main() {
6:     streampos begin, end;
7:     ifstream myfile("example.bin", ios::binary);
8:     begin = myfile.tellg();
9:     myfile. [ ] ;
10:    end = [ ] ;
11:    myfile.close();
12:    cout << "size is: " << (end-begin) << " bytes\n";
13:    return 0;
14: }

===== screen output =====
size is: 40 bytes

```

5. (20 points) Following C++ code contains one wrong part that will generate a compilation error.

Assume that the code from line 1 to line 7 is correct.

(a) Which line in the code is wrong? Mark the wrong part, where compilation error will occur, using an arrow and a circle.
 (b) What is wrong with the part of code you marked? (Why is it wrong?) Explain. ()

(c) Insert one line code in line 12 to make the code correct. ()

(d) What will be the screen output (i.e. execution result generated by **cout**) of the program after the correction you made?
 (screen output (execution result):)

```

1 : #include <iostream>
2 : using namespace std;
3 :
4 : class X {
5 : private:
6 :     int w,h;
7 : public:
8 :     X() { cout << "c1 called\n"; }
9 :     X(int a,int b) { w=a; h=b; cout << "c3 called\n"; }
10:    X(const X& x) { w=x.w; h=x.h; cout << "c5 called\n"; }
11:    ~X() { cout << "d called\n"; }
12:
13: };

```

```

15: int f1(X obj)
16: {
17:     cout << "f1 called!\n";
18:     return obj.w*obj.h;
19: }
20: int main()
21: {
22:     int z;
23:     X a(5,6) , b;
24:     X c = a;
25:     b = c;
26:     z = f1(b) ;
27:     cout << "area: " << z << "\n";
28:     return 0;
29: }

```